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CSC 450

May 9th, 2025

Assignment #9

1- What is the difference among deadlock avoidance, detection, and prevention?

Deadlock avoidance is a strategy that will “test” for deadlock by having allocation follow through, if the “test” fails, the allocation does not happen as it is regarded unsafe. If it succeeds, it moves to the safe state. Moreover, deadlock detection is more dangerous as it will let itself get into a deadlock. But, if it does, it takes the procedures needed to recover from it. With deadlock prevention, algorithms potentially prevent the deadlock state before allocation by eliminating one of the few conditions such as hold-and-wait or circular wait from occurring. Overall, they differ by avoidance requiring future knowledge on allocation of resources, detection differs with being more dangerous than the others by allowing unsafe states, and prevention has the potential to under utilize resources.

2- Can a system detect that some of its processes are starving? Explain your answer

A system cannot detect if a process is starving because it doesn't view the process in that regard. To an OS, it sees that other processes require more resources than the process being starved, deterring it. However, if times for each processes are tracked with flags, I believe each process will run because after a flag is met, the next process will begin.

3- Given the following state for the Banker's Algorithm, Can all of the four processes be run to completion with the resources available? If Yes, Show that the current state is safe.

5 processes and 3 resource type x (10 instances), y (5 instances), and z (7 instances).

Available Vector:

x	y	z
3	3	2

Allocation Matrix:

	x	y	z
P0	0	1	0
P1	2	0	0
P2	3	0	2
P3	2	1	1
P4	0	0	2

Claim Matrix:

x	y	z
7	5	3
3	2	2
9	0	2
2	2	2
4	3	3

Because P3's need vector does not exceed the allocation matrix, I will use that first to begin testing. $[3,3,2] + [2,1,1] = [5,4,3]$. Now, the available matrix is $[5,4,3]$ and does not exceed P1's need matrix. $[5,4,3] + [2,0,0] = [7,4,3]$. So far, P3 and P1 both are true. Next, I will run P4 since it does not exceed $[7,4,3]$. $[7,4,3] + [4,3,1] = [11,7,4]$. P4, then is true. P2 will now be run because it doesn't exceed the available vector. $[11,7,4] + [6,0,0] = [17,7,4]$. P2 is true. Finally, P0 will be tested as it definitely doesn't exceed the available vector now. $[17,7,4] + [7,5,3] = [24,11,7]$. Thus, the system is safe using the sequence P3,P1,P2,P4,P0.